

Module 1: Sourcing & Screening

Behavioral Finance - PE/VC Curriculum

Topic 2: Expected Utility Theory (EUT)

Topic 2: Expected Utility Theory (EUT) Formulation

We have established that if an agent obeys the von Neumann-Morgenstern axioms, their preferences can be represented by a utility function $U(w)$, where w is terminal wealth.

In an uncertain world, the agent does not maximize expected wealth; they maximize **Expected Utility**. For a discrete set of outcomes, the fundamental formula is:

$$E[U(w)] = \sum_{i=1}^n p_i \cdot U(w_i) \quad (1)$$

Where p_i is the probability of state i , and w_i is the wealth in that state.

Utility Curve Curvature (The Three Risk Profiles)

The *shape* of the utility curve $U(w)$ determines the investor's attitude toward risk. We assess this via the second derivative $U''(w)$:

- **Risk Aversion (Concave):** $U''(w) < 0$.
 - Diminishing marginal utility of wealth.
 - The pain of losing \$1M exceeds the joy of gaining \$1M.
 - *Default assumption for institutional LPs and GPs.*
- **Risk Neutrality (Linear):** $U''(w) = 0$.
 - Constant marginal utility.
 - The agent only cares about the arithmetic mean payoff.
- **Risk Seeking (Convex):** $U''(w) > 0$.
 - Increasing marginal utility.
 - The agent gambles for the thrill of large upside.

Jensen's Inequality (The Heavyweight Concept)

Jensen's Inequality is the mathematical shield of risk aversion. It states that for a concave function (i.e., a risk-averse utility curve), the utility of the *expected value* is strictly greater than the *expected utility* of the gamble.

$$U(E[w]) > E[U(w)] \quad \text{for concave } U(w) \quad (2)$$

Intuition: A risk-averse investor prefers a guaranteed \$10M exit over a 50/50 gamble that pays either \$0 or \$20M, even though both have an *expected value* of \$10M. The mathematical gap represents the psychic "cost" of bearing volatility.